Seohyun Park

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Education

Georgia Institute of Technology Atlanta, GA, USA
Bachelor of Science in Computer Engineering, *GPA 4.0/4.0* *August 2024 – May 2027 (Expected)*

- **Relevant Coursework:** Circuit Analysis, Programming for HW/SW Systems, Data Structures and Algorithms, Architecture, Systems, Concurrency and Energy in Computation, VLSI and Advanced Digital Design

Technical Skills

EDA & Digital Design: Cadence Innovus, Cadence Virtuoso (RTL-to-GDSII Certified), RTL (VHDL/SystemVerilog), Quartus Prime
Programming & Software: Tcl, Python, C/C++, Java, RISC-V Assembly, MATLAB, Git, Linux, Atlassian Jira
Hardware & Tools: Altera FPGA, oscilloscope, logic analyzer, multimeter

Experience & Activities

Apple — Silicon Engineering Group San Jose, CA
SoC Physical Design Engineering Intern *May 2026 – August 2026*

- Built a fully autonomous Python + Tcl orchestrator that anchors PnR stages (floorplan, macro placement, CTS, routing, signoff) on a reference partition for netlist reuse across die-shape and IO-pin perturbations.
- Designed a per-stage QoR gate (deterministic threshold ladder + LLM decision layer) and cross-stage rollback logic that decide when a stage continues, iterates, or rewinds against WNS/TNS setup/hold, DRC, congestion, and power targets.
- Validated across aspect-ratio reshape and IO-pin shift experiments on 3 production partition families with 5 completed full flows to signoff handoff, matching reference QoR (WNS/TNS, DRC, congestion) on all completed flows.

Georgia Tech — SiliconJackets (Chip Design Club) Atlanta, GA
Physical Design Engineer *August 2024 – May 2026*

- Designed and optimized fetch block microarchitecture within a 6-stage RISC-V core, reducing area inefficiencies and improving estimated frequency by ~5%.
- Worked within full RTL-to-GDSII flow (Cadence synthesis, place & route, signoff) on SkyWater 130nm process, reducing design rule violations from ~2000 to ~1000 on the Caravel SoC harness.

Emory Department of Biochemistry — Liang Lab Atlanta, GA
Computational Researcher *August 2022 – May 2026*

- Engineered a high-performance distributed pipeline using OpenMPI and Python to parallelize docking of 1.9M+ molecules across a 192-core cluster, reducing runtime from 47 seconds per molecule to 2 molecules per second.
- Automated data preprocessing and iterative filtering with OpenBabel, improving throughput and reproducibility of large-scale screening workflows, reducing computation time by 50%.

Projects

5-Stage Pipelined RISC-V Processor | *Georgia Tech Course: Computer Architecture* *May 2025 – July 2025*

- Implemented a 5-stage pipelined RV32I processor in SystemVerilog with hazard detection, stalling, data forwarding, and pipeline flushing; validated using ModelSim and benchmark programs.
- Built on a prior single-cycle RV32I design supporting ADD, ADDI, SUB, LW, SW, SLT, and JAL, with integrated instruction/data memory.

FPGA Morse Code LED Controller | *Georgia Tech Course: Digital Design Laboratory* *January 2025 – May 2025*

- Programmed a VHDL-based LED peripheral on the DE10 FPGA that converts ASCII input into Morse code patterns and displays them via LEDs and a 7-segment display.
- Implemented static and flashing modes via state machine design to interface with an SCOMP processor without altering internal architecture.

Snake Game on Mbed Platform | *Georgia Tech Course: Programming for Hardware/Software* *January 2025 – May 2025*

- Developed an interactive Snake game in C++, integrating hardware elements (Mbed microcontroller, nav-switch input, uLCD display), with game features including score tracking, win/lose states, and real-time collision detection.